

Project Report: Local Data Analysis of System Logs

1. Project Overview & Repository

The goal of this project is to extract, process, and visualize data from local Linux system log files to monitor system health and activity patterns.

- **GitHub Repository:** https://github.com/Leo-PHAV/EUTech_Leo-PHAV.git
- **Branch:** [task-log-analysis-Leo-PHAV](#)

2. Methodology & Technical Stack

- **Language:** Python 3.x
- **Libraries:** **Pandas** for data manipulation, **Matplotlib** and **Seaborn** for visualizations.
- **Parsing Technique:** Regular Expressions (Regex) were used to extract timestamps, hostnames, services, and log levels from raw text files.

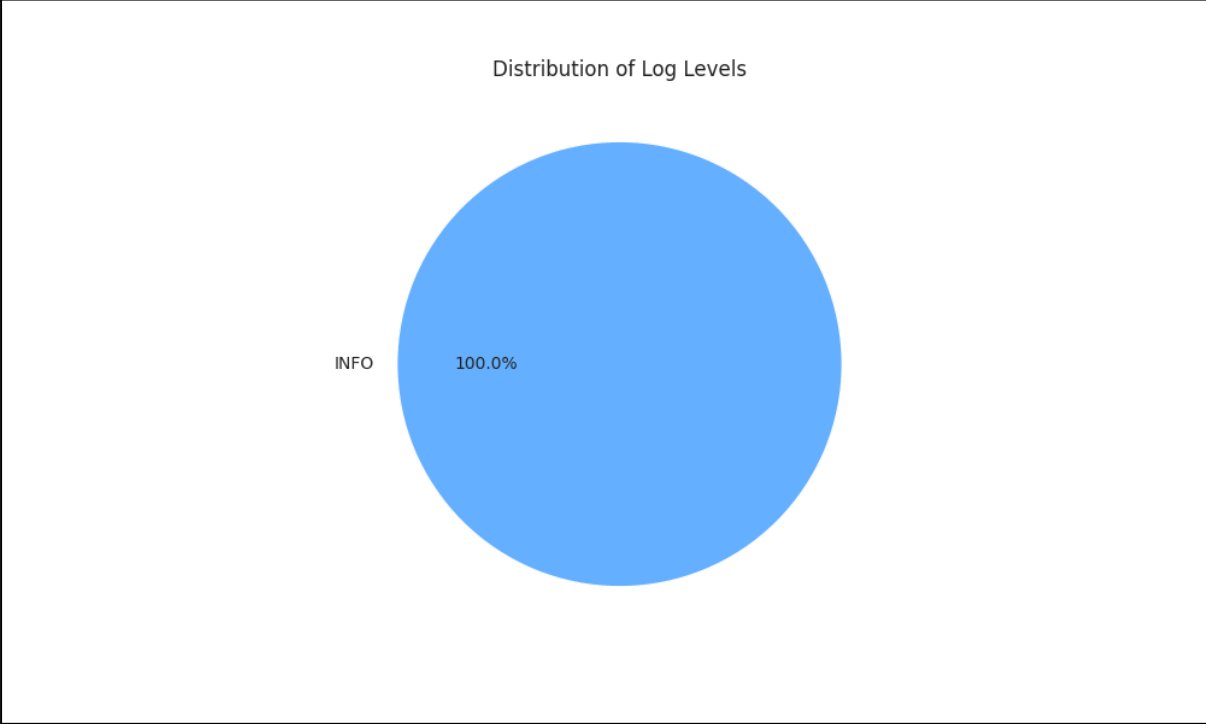
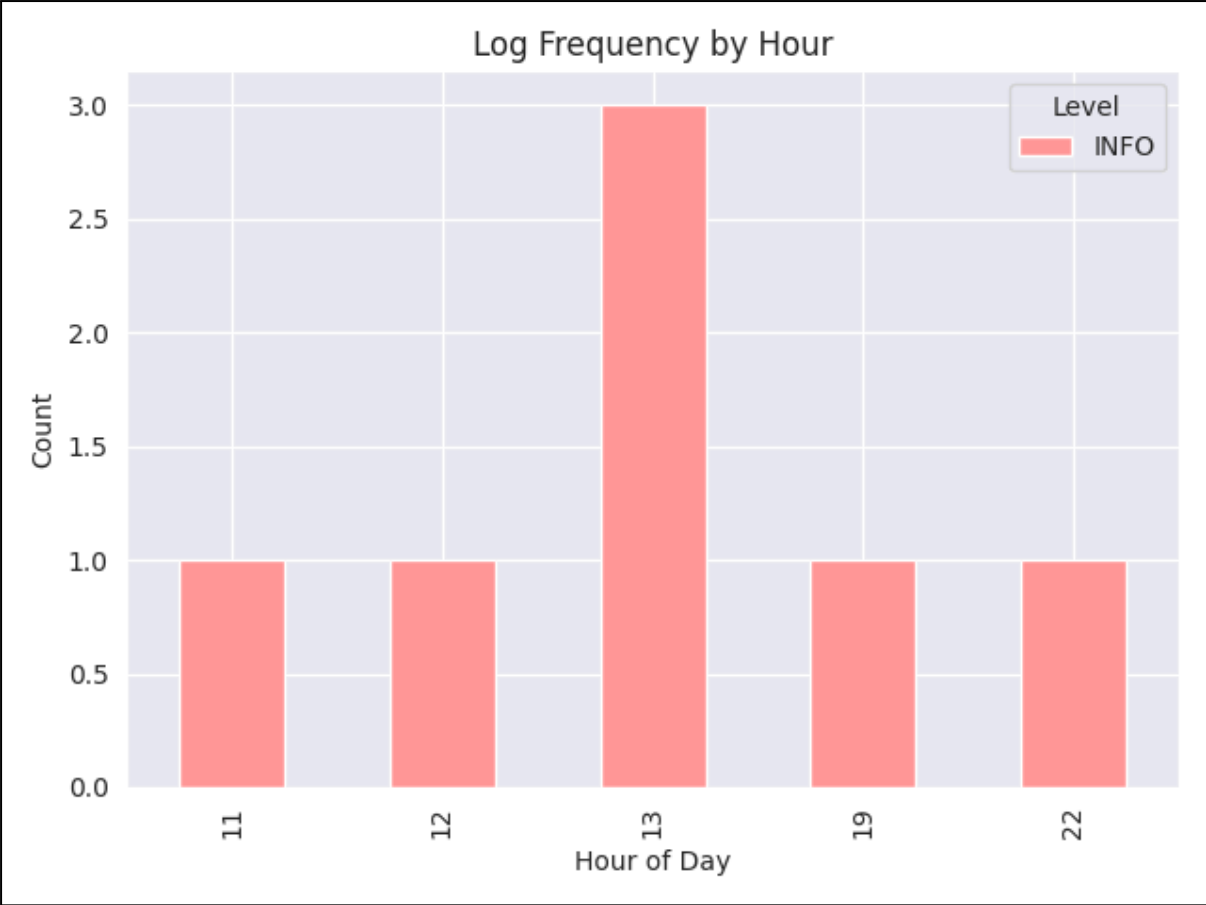
3. Log Identification

The following relevant log files were identified and analyzed to ensure a comprehensive overview:

- [/var/log/syslog](#): General system messages.
- [/var/log/auth.log](#): Security and user login events.
- [/var/log/kern.log](#): Kernel-level and hardware events.

4. Data Visualizations

- **Log Level Distribution:** A pie chart showing the ratio of **INFO** vs **ERROR** messages to identify system stability.
- **Hourly Activity Spikes:** A bar chart visualizing log frequency per hour to detect resource usage spikes or unusual activity periods.



5. Automation & Export

- **Export:** The script automatically generates an **HTML report** and a **summary.txt** file for easy auditing.
- **Automation:** The script is configured to run weekly via a **Cron job**.
 - *Command:* `0 8 * * 1 /path/to/python /path/to/src/log_analyzer.py`
 - This ensures a weekly log analysis without manual intervention.